

Estimating the Causal Effect of Higher Education on Antenatal Care (ANC) Utilization with Sensitivity Analysis Using Machine Learning Methods

Abu Sufiun Rhyme

Roll: FH-1033

02-03-2026

Institute of Statistical Research and Training
University of Dhaka

1. Background of the study
2. Objectives of the study
3. Methodology
4. Simulation
5. Time Frame

Background of the study

Background and Motivation

- Maternal education is an important determinant of maternal healthcare utilization in Bangladesh ([Akter et al., 2024](#)).
- Previous studies have found that women with secondary or higher education are more likely to receive adequate ANC (at least four visits) ([Akter et al., 2024](#)).
- Recent research has applied Causal Forests to estimate the causal effect of maternal education on adequate ANC utilization using BDHS 2022 data ([Mahmud et al., 2025](#)).
- However, causal estimates from observational data may remain sensitive to unmeasured confounding.
- Therefore, machine learning-based causal methods combined with sensitivity analysis can provide more robust evidence on the causal effect of higher education on ANC utilization.

Background and Motivation

- causal models (e.g., regression with propensity scores):
 - Assume prespecified functional forms, mostly linear/additive cause-effect relationship.
- May miss nonlinearities and complex interactions, leading to biased estimates.
- Machine learning algorithms (e.g., causal forests, BART, causal boosting) allow ([Wager and Athey, 2018](#)):
 - Flexible modeling without pre-specified forms
 - Capturing high-dimensional, nonlinear interactions
 - Estimating heterogeneous treatment effects
- In Bangladesh, the causal pathway from maternal High education to ANC utilization may involve complex interactions between socio-demographic and parental characteristics.

Background and Motivation

- Observational data may still have unmeasured confounders (e.g., cultural norms, maternal autonomy, household decision-making).
- Such unmeasured factors could bias causal effect estimates even after adjusting for observed variables.
- Machine learning algorithms improve modeling but cannot account for unmeasured confounding.
- Sensitivity analysis evaluates how strong an unobserved confounder must be to alter the estimated effect ([VanderWeele and Ding, 2017](#)).
- Enhances robustness, transparency, and credibility of causal conclusions.

Objectives of the study

Objectives of the Study

- Providing a comprehensive review of contemporary machine learning and Bayesian approaches for estimating causal effects of a binary treatment on a binary outcome.
- Making a systematic and thorough comparison of these machine learning and Bayesian methods with classical parametric model using a simulation study based on BDHS 2022 database.
- Suggesting a reliable method for estimating the causal effect of maternal early marriage on child stunting.
- Assessing bias from unmeasured confounders and evaluate robustness of causal estimates.

Methodology

Potential outcome framework for CATE

- The conditional average treatment effect (CATE) is an effective estimand to assess HTE. Within the potential outcomes framework (Holland, 1986; Rubin, 1974), CATE can be formally defined as follows.
- Consider a sample of n individuals indexed by $i = 1, \dots, n$. For each individual i , let X_i denote the vector of baseline covariates, $Z_i \in \{0, 1\}$ the binary treatment indicator, and $Y_i \in \{0, 1\}$ the binary outcome.
- Also consider the two potential outcomes for individual i : $Y_i(1)$ under treatment ($Z_i = 1$) and $Y_i(0)$ under control ($Z_i = 0$). The individual treatment effect is defined as

$$\tau_i = Y_i(1) - Y_i(0).$$

Potential outcome framework for CATE

- But only one of the two potential outcomes is observable for any given individual. Thus, the observed outcome is

$$Y_i = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

- Consequently, we cannot observe the treatment effect at the individual level, τ_i .
- Within this framework, we assume conditional exchangeability, $\{Y(1), Y(0)\} \perp\!\!\!\perp Z \mid X$, and positivity, $0 < P[Z = 1 \mid X = x] < 1$.
- The term $P[Z = 1 \mid X = x]$ represents the propensity score, $e(x)$, which is the probability of receiving the treatment given the baseline covariates.

Potential outcome framework for CATE

- Under these assumptions, the conditional mean functions can be defined as:

$$\mu_1(x) = \mathbb{E}[Y_i \mid X_i = x, Z_i = 1], \quad \mu_0(x) = \mathbb{E}[Y_i \mid X_i = x, Z_i = 0].$$

- The conditional average treatment effect (CATE) can then be written as:

$$\tau(x) = \mu_1(x) - \mu_0(x).$$

- The average treatment effect (ATE), $\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$, can be obtained by taking the expectation of $\tau(x)$ over the distribution of the relevant population.

Machine Learning Approaches

- Regression of the conditional mean functions using flexible, nonparametric techniques derived from standard machine learning models can efficiently capture heterogeneous treatment effects (HTE) ([Wendling et al., 2018](#)).
- In this study, we consider two distinct methods for modeling the conditional mean:
 1. G-computation: Bayesian additive regression trees (BART).
 2. Causal learning: Neural Networks, Causal Trees, Causal Forests, and Causal Boosting.
- Inverse Probability Weighting (IPW) weighting is employed as the reference (benchmark) approach.

Evaluation criteria

- The model's performance will be evaluated using the root mean squared error (RMSE) of the individual-level CATE estimates, $\hat{\tau}_i(x)$, calculated as

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i(x) - \tau_i(x))^2}.$$

- Additionally, the performance of the ATE estimate (marginal risk difference) will be assessed using the dataset. This is quantified by the absolute relative error, expressed as

$$\text{Absolute Relative Error} = 100 \times \frac{|\hat{\tau} - \tau|}{\tau},$$

where $\hat{\tau}$ and τ represent the estimated and true ATE, respectively.

Simulation

Simulation experiments

- In this study, the simulation procedure is based on empirical data, utilizing observed covariates and treatment assignments directly.
- Only the outcomes are generated using nonparametric models based on the observed data.
- In the simulated datasets, we apply the specified methods to estimate CATE and evaluate their performance.

- The Bangladesh demographic health survey (BDHS) 2022 data will be used in our study.
- The outcome of interest is ANC utilization, which is binary.
- The exposure of interest is higher education , which is also binary.

- Maternal higher education is defined as having completed secondary or higher education, while women with primary or no formal education are classified as having lower educational attainment. ([BDHS, 2022](#)).
- Women who received at least four antenatal care (ANC) visits are classified as having adequate ANC utilization (1), while those who received fewer than four visits are classified as having inadequate ANC utilization (0) ([BDHS, 2022](#); [World Health Organization, 2024](#)).

Data generating process

- We will follow ([Wendling et al., 2018](#)) to set up the procedure for data generation.
- We will use the conditional risk difference as the CATE estimand. The treatment effect function is defined as:

$$\tau(x) = \pi^1(x) - \pi^0(x),$$

where,

$$\pi^1(x) = P[Y = 1 \mid X = x, Z = 1]$$

$$\pi^0(x) = P[Y = 1 \mid X = x, Z = 0],$$

represent the probabilities of the potential outcome given treatment and control, respectively.

Data generating process

1. We will model $\pi_a(x)$ using a non-parametric model (e.g., neural network) and estimate the conditional probabilities $\pi_1(x)$ (treatment) and $\pi_0(x)$ (control) from the fitted model.
2. Then we define the treatment effect function on the linear predictor scale of a logistic regression model as:

$$\gamma(x) = \text{logit}[\pi^1(x)] - \text{logit}[\pi^0(x)] + c,$$

Here, c is a constant that is determined iteratively to ensure that the mean of $\gamma(x)$ matches the logarithm of the odds estimated from the original observational study.

3. The probability of the potential outcome under treatment is recalculated as

$$\pi_{\text{adjusted}}^1(x) = \text{logit}^{-1}[\text{logit}[\pi^0(x)] + \gamma(x)],$$

and the treatment effect, expressed on the probability scale as the risk difference, is calculated using

$$\tau(x) = \pi_{\text{adjusted}}^1(x) - \pi^0(x)$$

4. The factual outcome is drawn as

$$Y \sim \text{Bernoulli}(\pi_{\text{factual}}(x)), \text{ with } \pi_{\text{adjusted}} = \pi_{\text{adjusted}}^1(x)Z + \pi^0(x)(1-Z)$$

Data generating process

- A standard logistic regression framework (Step 3) will be used to generate the outcome under the treatment effect.
- Steps 2 and 3 allow us to fix the mean of the treatment effect function $\tau(x)$ to a desired value based on a marginal effect estimate.
- Each simulated dataset will include X , Z , Y^{sim} , $\pi_{\text{factual}}(x)$, and $\tau(x)$, which are saved for evaluating model performance.

Sensitivity Analysis for Unmeasured Confounding

- If the unmeasured confounder is collinear with a measured one, adjusting for it has little effect on the causal estimate ([Groenwold et al., 2010](#)). Therefore, we assume the unmeasured confounder is not collinear with measured confounders.
- The simulation design follows the approach proposed by [Groenwold et al. \(2010\)](#) which assumes a binary unmeasured confounder.
- The impact of unmeasured confounding is evaluated by varying the assumed prevalence of the unmeasured confounder, as well as the strength of its association with the exposure and with the outcome.

Time Frame

Time Frame

Time	Task
January, 2026	Idea formulation and literature review
February, 2026	Preparation and presentation of the thesis proposal
June, 2026	Methodology development
July, 2026	Coding
August, 2026	Computation and simulation
September, 2026	Completion of the first draft of the thesis report
October, 2026	Final draft completion, submission, and thesis defense

- Akter, E., Rahman, M. S., Hossain, A. T., Siddique, A. B., Sayeed, A., Chowdhury, P. B., El Arifeen, S., Rahman, A. E., and Ahmed, A. (2024). Evaluation of the causal effect of maternal education on the utilisation of maternal health services in bangladesh using an observational study – a comparison of different propensity score methods and covariate adjustment. *Archives of Public Health*, 82(207).
- BDHS (2022). Bangladesh demographic and health survey 2022: Key indicators report (pr148).
<https://www.dhsprogram.com/pubs/pdf/PR148/PR148.pdf>.
Accessed: 2026-02-21.
- Groenwold, R. H., Nelson, D. B., Nichol, K. L., Hoes, A. W., and Hak, E. (2010). Sensitivity analyses to estimate the potential impact of unmeasured confounding in causal research. *International journal of epidemiology*, 39(1):107–117.

- Holland, P. W. (1986). Statistics and causal inference. *Journal of the American statistical Association*, 81(396):945–960.
- Mahmud, S., Mohsin, M., Mustani, R., and Akter, M. S. (2025). Integrating machine learning for propensity score matching and causal inference: A causal forest approach to assessing the impact of maternal education on antenatal care utilization. *medRxiv*.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of educational Psychology*, 66(5):688.
- VanderWeele, T. J. and Ding, P. (2017). Sensitivity analysis in observational research: introducing the e-value. *Annals of internal medicine*, 167(4):268–274.
- Wager, S. and Athey, S. (2018). Estimation and inference of heterogeneous treatment effects using random forests. *Journal of the American Statistical Association*, 113(523):1228–1242.

- Wendling, T., Jung, K., Callahan, A., Schuler, A., Shah, N. H., and Gallego, B. (2018). Comparing methods for estimation of heterogeneous treatment effects using observational data from health care databases. *Statistics in medicine*, 37(23):3309–3324.
- World Health Organization (2024). Malnutrition. <https://www.who.int/news-room/fact-sheets/detail/malnutrition>. Accessed: 2026-02-21.

Thank You!